MILITARY SPECIFICATION

CONTROL, AIRCRAFT WEAPONS RELEASE SET

C-8014(*)/AWE-1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall enable the pilot to select the quantity of stores to be dropped, mode of release, and drop interval between release of individual stores.

1.2 Classification - Aircraft weapons release set control C-8014(*)/AWE-1 shall consist of the item listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.4).

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(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including transistors and diodes) shall be obtained as outlined in Specification MIL-E-5400.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 22 ounces.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 10,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 $\pm 5^{o}$C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 $\pm$ 1.0 vac and/or 27.5 $\pm$ 0.5 vdc

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for class II equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of curve I of Specification MIL-E-5400.

3.3.10.2 Altitude - The equipment shall operate satisfactorily when subjected to the altitude requirements of Specification MIL-E-5400 with the exception that the operational altitude be limited to 50,000 feet.

3.3.11 Warm-Up Time - The equipment shall require no warm-up time.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. (See figure 1.)

(1) AC Power (Single Phase) 115V, 400Hz, Category "B", 10 VA

(2) AC Power (Single Phase) 5V, 400Hz, Category "B", 5 VA

(3) DC Power 28V, Category "B", 0.25 amp, 7 w

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3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The control is a manually operated unit from which the pilot makes all his selections as to pertinent drop information.

3.4.2 Types of Release - The types of release which can be selected by the unit are divided into two basic catagories as follows:

3.4.2.1 Ripple - In any ripple type of release, stores release will automatically continue at a selected drop interval as described in 3.4.4, until the pilot has released the desired quantity of stores selected as described in 3.4.5.

3.4.2.2 Step - In any step type of release, stores release will occur at a rate determined by a pilot. This rate will be one release pulse generated for each pilot depression of the weapon release button.

3.4.3 Modes - The modes of release are as follows:

3.4.3.1 Singles - One store will be released for each output pulse.

3.4.3.2 Pairs - Two stores will be released for each output pulse.

3.4.3.3 Salvo - The number of stores released for each output pulse will be equal to the number of loaded stores stations selected.

3.4.4 Rate of Release - The interval between successive release output pulses in any ripple mode shall be pilot controlled and continuously variable between 20 and 2000 milliseconds. The output pulse interval shall be within ±5 percent of the control setting for standard conditions and ±10 percent for extreme conditions.

3.4.4.1 Pulse Width - The duty cycle of the output pulses shall be 50 percent of the output pulse interval described in 3.4.4 with accuracies as follows:

Pulse Interval	Accuracy
20 - 50 ms	±40 pct
50 - 2000 ms	±10 pct

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In the step type of release, the pulse width shall be determined by the pilot. The leading edge of the release pulse shall occur when the pilot manually starts a step drop. The trailing edge of the pulse shall occur when the pilot manually ends the drop.

3.4.5 Quantity Select - The number of release output pulses generated by the system shall be pilot controlled. The pilot, when desiring a drop of any amount of stores, shall be able to select the exact amount of from 2 to 40 output pulses to be generated in amounts limited to the following steps:

(1) 1, 2, 3, 4, 5, 6 (steps of one)

(2) 8, 12, 16, 20 (steps of four)

(3) 30, 40 (steps of ten)

3.5 Detailed Requirements -

3.5.1 Control, Aircraft Weapons Release Set C-8014(*)/AWE-1 - The control shall meet the following requirements:

3.5.1.1 Function - The control is the unit of the system which selects the operating modes of the system. The control shall determine the following:

(1) The quantity of stores to be released, as tabulated in 3.4.5.

(2) The drop interval which shall range from 20 to 2000 milliseconds.

(3) The mode and type of release as tabulated in 3.4.2.

3.5.1.2 Form Factor - The control shall conform to the dimensions and requirements shown in Figure 2. It shall be mounted in the cockpit.

3.5.1.3 Weight - The weight of the control shall not exceed 1.25 pounds.

3.5.1.4 Contents - The control shall contain the electronics to provide an external intra-system 28 volt pulse train whose frequency shall be controlled by the drop interval select knob of 3.5.1.5. The width of this pulse shall be as specified in 3.4.4.1.

3.5.1.5 Controls - All controls and select switches are located and identified as in Figure 2. The controls and switches perform the following functions.

Reference Designation	Function
QTY - SEL	Quantity select and system turn on
DROP INTVL	Drop interval in milliseconds
MULT SW	Drop interval multiplier switch
MODE SELECT	Determines mode of operation

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3.5.1.6 Electrical Connections - Connections to external circuits shall be provided as indicated in figure 2.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see 6.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction tests

(2) Initial production tests

(3) Acceptance tests

(4) Life tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approval test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for reviews and approval. On first production contracts, this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating conditions.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Tests Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities

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satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual tests

(2) Sampling tests

(3) Reliability assurance tests

(4) Special tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of product

(2) Operational test

(3) Manufacturing run in test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect

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workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless reliability assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to Be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the initial production test (4.3) is deleted, then a sample test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by MIL-STD-781. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Test Level - The test level shall be F from table I of MIL-STD-781.

4.4.3.2 Accept-Reject Criteria - Test plan V of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability qualification test plan. Test plan VI of MIL-STD-781 shall be used to determine the accept-reject criteria for the reliability sampling test plan.

4.4.3.3 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 2 hours during each cycle.

4.4.3.4 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.10.

4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests, no preventive maintenance measures are required.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to Be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room ambient
AC voltage	115 ±5 volts (at lowest applicable frequency)
DC voltage	27.0 ±2.0 volts

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) Circuit reference symbol number

(3) Part function

(4) Name of manufacturer

(5) Nature of failure

(6) Number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - Equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such

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procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The unit is used for control of Aircraft Weapons Release Set AN/AWE-1, Specification MIL-A-81359(AS).

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 $\pm$0.5 vdc and 115 $\pm$1.0 vac are applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held

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to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.9 Items of Equipment - The equipment covered by this specification shall consist of the following item:

Item	Type Designation	Appl. Para.
Control, Aircraft Weapons Release Set	C-8014(*)/AWE-1	3.5

6.9.1 Associated Equipment - The equipment shall operate with the following associated equipment.

Item	Type Designation	Specification
Programmer, Aircraft Weapons Release Set	MX-7016(*)/AWE-1	MIL-P-81361(AS)
Indicator, Aircraft Weapons Release Set	ID-1310(*)/AWE-1	MIL-A-81359(AS)

6.10 Performance Characteristics to Be Measured - The equipment shall meet the performance characteristics specified in tables I and II when connected in accordance with figure 1.

Table I

Control Positions				*Test Points	
Weapons Release Switch	Quantity Select	Drop Interval	Mode Select	Connector Pin	Output Readings
Open	"Off"	Any Setting	Ripple	1P1-c	<+100MV DC
				1P1-A	<+100MV DC
				1P1-C	<+100MV DC
				1P1-P	<+100MV DC
				1P1-g	<+100MV DC
				Polarity of resistance of CR15 between 1P1-A and 1P1-g shall be < 1K and > 999K, respectively	
Open	Two	Any Setting	Step Salvo	1P1-P	<+100MV DC
				1P1-C	<+100MV DC
				1P1-g	≥ +26 VDC

*All readings with respect to 28 vdc ground for common.

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Table I (Cont)

Control Positions				*Test Points	
Weapons Release Switch	Quantity Select	Drop Interval	Mode Select	Connector Pin	Output Readings
Open	Two	Any Setting	Ripple Salvo	1P1-P 1P1-C 1P1-g	< +100MV DC < +100MV DC ≥ +26 VDC
Close	Two	Any Setting	Ripple Salvo	1P1-w	> 5.0 VDC
Open	Two	Any Setting	Ripple Salvo	1P1-w	< +0.5 VDC
Open	Two	Any Setting	Step Single	1P1-P 1P1-C 1P1-g	≥ +26 VDC < +100MV DC < +100MV DC
Open	Two	Any Setting	Ripple Singles	1P1-P 1P1-C 1P1-g	≥ +26 VDC < +100MV DC < +100MV DC
Open	Two	Any Setting	Step Pairs	1P1-P 1P1-C 1P1-g	< +100 MV DC ≥ +26 VDC < +100MV DC
Open	Two	Any Setting	Ripple Pairs	1P1-P 1P1-C 1P1-g	< +100MV DC ≥ +26 VDC < +100MV DC

Pin H of 1P1 shall be chassis ground.
Pins H and C of 1P1 shall not be common.
*All readings with respect to 28 vdc ground for common.

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With the weapons release switch open, place the mode select switch at ripple salvo. Set the drop interval at any setting. With the quantity select switch in the positions indicated in table II, measure between the output pins on 1P1 and the 28V ground. The measured outputs shall be as shown in the table below.

Table II

"0" < +125 MV DC "1" ≥ 27 VDC

Quantity Select Switch Position	1P1 Output Pins											
	D	x	E	y	F	G	J	M	K	a	b	A
"off"	0	0	0	0	0	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	0	1
3	0	1	0	0	0	0	0	0	0	0	0	1
4	0	0	1	0	0	0	0	0	0	0	0	1
5	0	0	0	1	0	0	0	0	0	0	0	1
6	0	0	0	0	1	0	0	0	0	0	0	1
8	0	0	0	0	0	1	0	0	0	0	0	1
12	0	0	0	0	0	0	1	0	0	0	0	1
16	0	0	0	0	0	0	0	1	0	0	0	1
20	0	0	0	0	0	0	0	0	1	0	0	1
30	0	0	0	0	0	0	0	0	0	1	0	1
40	0	0	0	0	0	0	0	0	0	0	1	1

6.10.1 Drop Interval - The pulse output shall be +16 vdc or greater when the unit is operating in all ripple modes with the weapons release switch closed. The pulse output on 1P1-c when the unit is operating in all step modes with the weapons release switch closed shall be +27 vdc or greater and less than +125 millivolts when the weapons release switch is open.

6.11 Applicable Documents -

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution

PUBLICATIONS

Naval Air Systems Command

MIL-A-81359(AS)	Aircraft Weapons Release Set AN/AWE-1
MIL-P-81361(AS)	Programmer, Aircraft Weapons Release Set MX-7016 (*)/AWE-1

6.11.1 Availability of Documents -

(1) When requested specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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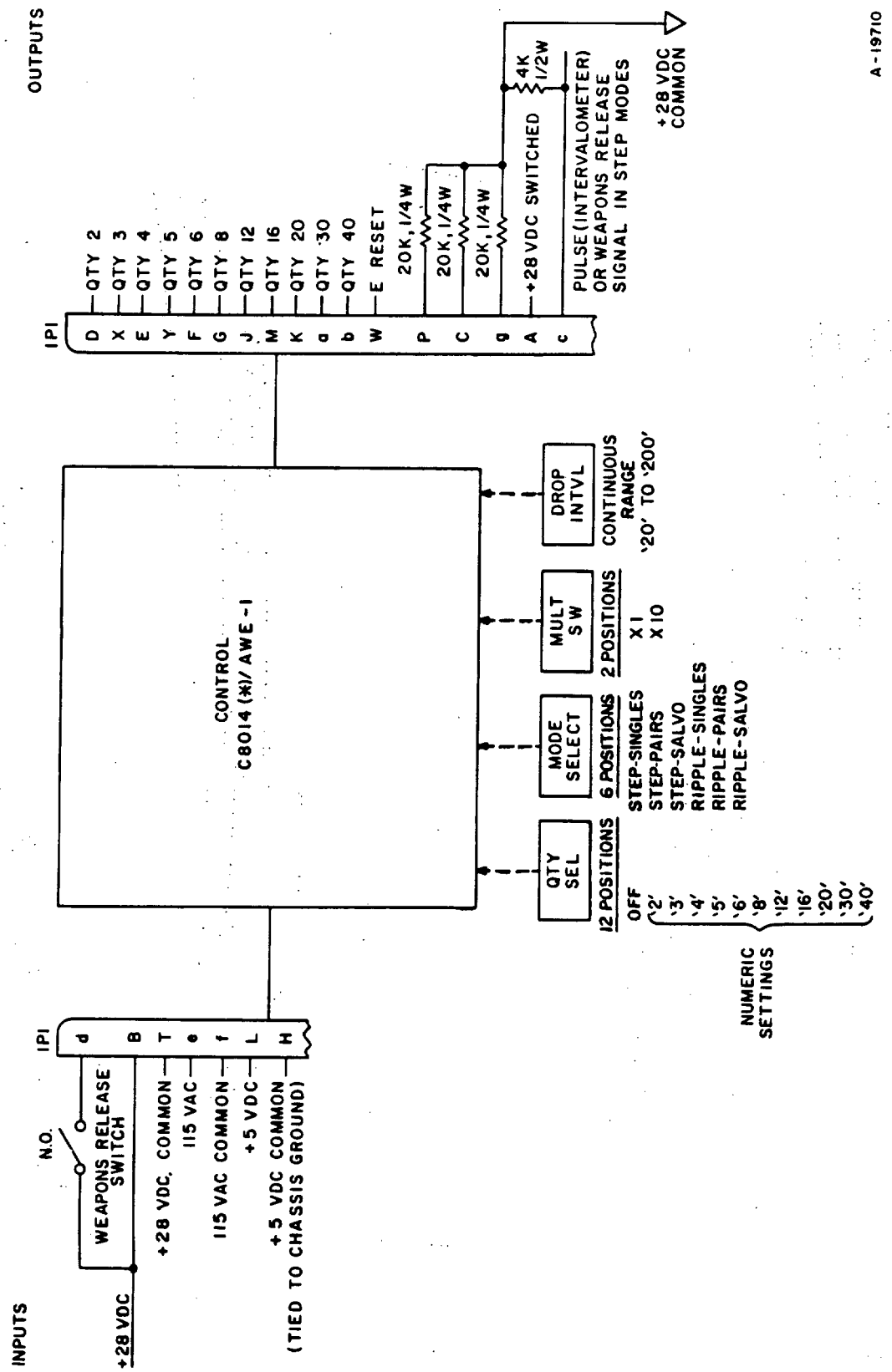


Figure 1. Operational Test Setup

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81770(AS) Control, Aircraft Weapons Release Set C-8014(*)/AWE-1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

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